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APPENDIX A, SECTION III

30. SPRING MANUFACTURE

30.1 GENERAL. Information on only a few operations is essential for design purposes.

1. STRESS RELIEVING. The usual types of hardening and tempering ovens are used for stress relieving. Springs made from pre-hardened wire such as Music Wire, Oil Tempered, Hard Drawn, Corrosion Resisting 18-8 and similar materials are stress relieved by heating at low temperatures from 400 to 650° F. to reduce the residual stresses trapped in the wire during the coiling operation. Springs made from annealed wire are hardened and tempered in a manner somewhat similar to tool steel. Precipitation hardening materials such as Beryllium-Copper, K-Monel, Inconel X, 17-7 PH and others are heated at varying temperatures, depending upon composition, for extended times from 1 hour to 16 hours.

2. COLD SET TO SOLID. This process is used to stabilize the free length of a compression spring, so that subsequent inadvertent, or intentional, compression to solid height will not change the loads at working deflections.

(a) If a compression spring is designed so that the elastic limit is not exceeded when the spring is compressed to solid height, no appreciable permanent set will occur, other than removal of small kinks in the wire. The note "Cold Set to Solid" should be specified on the drawing of such springs.

(b) If a spring is designed so that the elastic limit is exceeded when the spring is closed solid, permanent set will occur and the free length will be decreased. Residual stresses of opposite sign will be set up in the wire when the load is released, so that if the spring again is closed solid it will withstand a higher calculated stress than the stress corresponding with the elastic limit. If the initial free length of the spring is

made greater than the calculated free length by the proper amount, overstressing the spring beyond the elastic limit by compressing it to solid height will stabilize the characteristics and produce the desired loads at working deflections. Additional cycles of compressing the spring to solid height and releasing the load will not further change the free length. However, there is a limit to this process. After a certain initial free length has been reached for a particular spring, the final free length after compression to a solid height will remain constant no matter what increases are made in initial free length.

(c) When a spring is designed so that the stress at solid height is so far above the elastic limit that the spring will not have the desired loads at working deflections if cold set to solid, the note "Shall compress to in. without permanent set" should be placed on the drawing. The computed stress at the specified length (equal to, or less than, the final assembled length) must be less than the stress at the elastic limit. Whenever practicable, this design of spring should be discarded in favor of a spring having a solid stress within limits that will permit closing solid without permanent set.

Several methods may be used to cold set to solid. Small air presses and foot presses can be used for light springs, power presses, and specifically built motorized equipment for larger springs, and hydraulic presses for heavy springs. Light springs also may be compressed by hand over close fitting arbors.

33. GRINDING. End coils of compression springs are ground whenever it is necessary for the springs 1. to stand upright, 2. to obtain a good seat against a contacting part, 3. to reduce buckling, and 4. to cause the springs to exert more uniform pressures under a diaphragm or against a mating part. This operation is expensive and should be avoided wherever it is practicable to do so—especially on light springs with wire diameters

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under $\frac{1}{32}$ inch and where a large spring index ratio (OD/d) prevails, such as 13 or larger. Several types of disc grinding machines are available. The small disc grinders using paddles to hold springs for grinding are especially useful for small production runs from 100 to 5000 or more springs. Grinders with horizontal "ferris wheel" methods of loading are useful for larger quantities of light springs using wire diameters up to about $\frac{3}{32}$ inch. Heavier sizes may be ground in disc grinders with vertical ferris wheels.

34. SHOT PEENING. Spring life can be increased at least 30 per cent and has often been increased from two to ten times by shot peening. This process may be applied to all highly stressed springs made from steel and non-ferrous materials usually over $\frac{1}{16}$ inch wire diameter. Extension springs and closely wound torsion springs are difficult to shotpeen because the tiny steel shot is frequently trapped between the coils and is difficult to remove. The large increase in fatigue life of helical springs due to shot peening is accomplished by a combination of effects:

(a) Small surface irregularities seen only by the microscope are hammered smooth.

(b) The surface of the wire is thoroughly cleaned, and sharp burrs are made dull.

(c) This additional cold work hardens the surface of the wire and raises the physical properties where the stress is highest.

(d) Cold forging traps beneficial compression stresses near the wire surface which must be overcome by the destructive tensile stresses that cause fatigue failure before breakage can occur. All heat-treating of springs and all stress-relieving processes should be done prior to shot peening, except in those instances where electroplating is used; it is then necessary to reheat after plating. Heating the springs above 500° F. after shot peening counteracts much of the

beneficial effects of the trapped compression stresses produced by shot peening.

Springs made from annealed, oil tempered or alloy steels that must be electroplated can be shot peened principally to clean the surface, thus avoiding the necessity of soaking them in acid solutions to remove scale. The slightly roughened surface of shot peened springs does not produce a bright glossy, electroplated coating.

35. PROTECTIVE COATINGS. Uncoated or oil dipped springs are satisfactory where corrosive conditions are not a factor. Black jannanning is often used as it is a flexible, inexpensive finish suitable for many applications. Enamels, lacquers and paint are occasionally used. Cadmium with supplementary chromate treatment provides one of the best electro deposited coatings because it is both flexible and corrosion resistant.

Nickel, chromium, zinc, and other metals are also deposited electrolytically. Black oxide and phosphate coatings are occasionally specified. Care must be used in applying coatings, first to use detergents, solvents, alkaline or vaporizing degreasers to remove oil and dirt, shotblasting or sand blasting to remove scale, and baking to remove the hydrogen embrittlement that may be caused by acid dipping or electroplating.

36. HYDROGEN EMBRITTLEMENT. Steel, particularly hardened steel, is susceptible to embrittlement resulting from hydrogen introduced by acid pickling, electroplating, or cathodic electrocleaning operations. Absorbed hydrogen results in brittle behavior particularly under sustained loading in the presence of stress concentrations. Baking to relieve hydrogen embrittled springs should be accomplished by a method similar to that specified in specification QQ-Z-325 or QQ-P-416.

(a) Most hydrogen is evolved during acid pickling and can be reduced by using organic inhibitors in the solution. The cleaning of

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springs by shotblasting or by tumbling is advisable so that less time is required in the pickling bath.

(b) Cathodic electrocleaning, by making the work anode instead of cathode, reduces the amount of hydrogen evolved and shortens the time required in acid dipping. This procedure is seldom used for springs because of the long time required to perform this operation for large quantities.

(c) Hydrogen evolution during electroplating can be reduced by using low current densities. Low voltages (between 6 and 8 volts) may be used. Improper or speedup methods are harmful. Commercial electroplating of springs may be unsatisfactory, due to the high current densities commercially used which cause a greater evolution of hydrogen and, therefore, more embrittlement.

(d) All electroplated springs made from steel should be dehydrated as soon as possible after plating. A time delay of 5 hours may prove too long in some cases.

High carbon spring steel is more susceptible to hydrogen embrittlement than the low-carbon steels, and faulty heat-treatment prior to plating will produce a type of grain structure that tends to increase embrittlement. Springs should not be plated while in a stressed condition or when part of an assembly, because stressed conditions increase embrittlement. For this reason the stretching of extension springs, so that deposition will occur between the coils, is not recommended.

Dehydrogenation, or removing of hydrogen, restores the normal condition of the steel. It is accomplished by heating the electroplated springs in an oven and should be done as soon as the springs are plated and washed. Correct temperatures and length of time at heat depend upon the type and size of the steel. Recommended heat for air circulating ovens is $375^{\circ} \pm 25^{\circ} \text{ F}$ for not less than 3 hours after plating.

Occasional embrittlement may occur even with the best regulated methods. Nonferrous

TABLE XVI. Tolerances on outside diameter for all helical springs

Outside diameter (in.)	Plus or minus tolerance on OD		
	Spring index (OD/d)*		
	Up to 6 (small)	6 to 10 (average)	10 to 16 (large)
$\frac{1}{8}$ to $\frac{1}{4}$	0.002	0.003	0.004
$\frac{1}{4}$ to $\frac{3}{8}$	0.003	0.004	0.006
$\frac{3}{8}$ to $\frac{1}{2}$	0.004	0.006	0.008
$\frac{1}{2}$ to $\frac{3}{4}$	0.006	0.008	0.010
$\frac{3}{4}$ to 1	0.008	0.010	0.012
1 to $1\frac{1}{2}$	0.010	0.012	0.015
$1\frac{1}{2}$ to 2	0.012	0.015	0.022
2 to 3	0.015	0.022	0.031
$3\frac{1}{2}$ to 4	0.022	0.031	0.047
4 to 5	0.047	0.062	0.094
5 to 6	0.062	0.094	0.125
6 to 8	0.094	0.125	0.156

* The spring index is based on OD for convenience.

TABLE XVII. Tolerances on the number of coils for all helical springs

Compression	Extension		Torsion
In order to meet the requirements of load, rate, free length, solid height, it is necessary to vary the number of coils by plus or minus: 5 per cent	Number of coils	Tolerance plus or minus	Same as extension wherever the design permits. Otherwise the number of coils will vary in accordance with the position of the ends as shown in following table.
	3 to 5	20°	
	6 to 8	30°	
	9 to 12	40°	
	For each additional coil add $1\frac{1}{2}^{\circ}$ per coil.		

Closer tolerances require trimming after coiling and increase manufacturing time and cost.

TABLE XVIII. Tolerances on position of arms for torsion springs (For indexes OD/d up to 10)

Total coils	Tolerance plus or minus
Up to 3	8°
Over 3 including 10	10°
Over 10 including 20	15°
Over 20 including 30	20°
Over 30	25°

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TABLE XIX. Tolerances on free length for compression springs with ends closed and ground and for accurately made extension springs

Length springs, in.	Total number of coils	Plus or minus tolerance on length, in.		
		Spring index (O.D. ÷ d)		
		Up to 4 (small)	6 to 10 (average)	10 to 16 (large)
Up to ¾, inc.	Up to 4	0.007	<i>0.011</i>	0.015
	Over 4 to 7	0.011	0.015	0.018
	Over 7 to 11	0.015	0.018	0.022
	Over 11	0.018	0.022	0.026
Over ¾ to 1½, inc.	Up to 5	0.010	<i>0.015</i>	0.020
	Over 5 to 10	0.015	0.020	0.025
	Over 10 to 15	0.020	0.025	0.030
	Over 15	0.025	0.030	0.035
Over 1½ to 2½, inc.	Up to 10	0.020	<i>0.030</i>	0.040
	Over 10 to 20, inc.	0.020	0.040	0.050
	Over 20 to 30, inc.	0.040	0.050	0.060
	Over 30	0.050	0.060	0.070
Over 2½ to 3½, inc.	Up to 15	0.030	<i>0.045</i>	0.060
	Over 15 to 30	0.045	0.060	0.075
	Over 30 to 45	0.060	0.075	0.090
	Over 45	0.075	0.090	0.105
Over 3½ to 4½, inc.	Up to 20	0.040	<i>0.060</i>	0.080
	Over 20 to 40	0.060	0.080	0.100
	Over 40 to 60	0.080	0.100	0.120
	Over 60	0.100	0.120	0.140
Over 4½ to 6, inc.	Up to 25		<i>0.075</i>	0.100
	Over 25 to 50	0.075	0.100	0.125
	Over 50 to 75	0.100	0.125	0.150
	Over 75	0.125	0.150	0.175
Over 6 to 8, inc.	Up to 35	0.070	<i>0.105</i>	0.140
	Over 35 to 70	0.105	0.140	0.175
	Over 70 to 100	0.140	0.175	0.210
	Over 100	0.175	0.210	0.245
Over 8 to 10, inc.	Up to 45	0.090	<i>0.135</i>	0.180
	Over 45 to 90	0.135	0.180	0.225
	Over 90 to 135	0.180	0.225	0.270
	Over 135	0.225	0.270	0.315
Over 10 to 14, inc.	Up to 60	0.120	<i>0.180</i>	0.240
	Over 60 to 120	0.180	0.240	0.300
	Over 120 to 180	0.240	0.300	0.360
	Over 180	0.300	0.360	0.420
Over 14 to 18, inc.	Up to 80	0.160	<i>0.240</i>	0.320
	Over 80 to 160	0.240	0.320	0.400
	Over 160 to 240	0.320	0.400	0.480
	Over 240	0.400	0.480	0.560
Over 18 to 22, inc.	Up to 100	0.200	<i>0.300</i>	0.400
	Over 100 to 200	0.300	0.400	0.500
	Over 200 to 300	0.400	0.500	0.600
	Over 300	0.500	0.600	0.700
Over 22 to 28, inc.	Up to 125	0.250	<i>0.375</i>	0.500
	Over 125 to 250	0.375	0.500	0.625
	Over 250 to 375	0.500	0.625	0.750
	Over 375	0.625	0.750	0.875

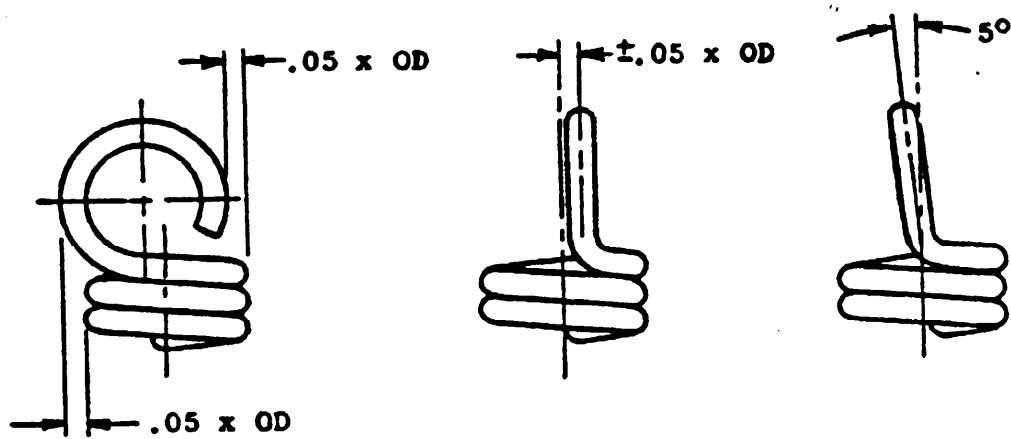
1. These tolerances may be reduced 50 percent by individually grinding, measuring, testing and sorting the springs, but this procedure increases production time and cost.

2. These tolerances should be doubled for compression springs that do not have ground ends and for the usual run of commercial extension springs.

3. The values in *italics* are standard when the wire diameter or number of coils is not predetermined, or where there is doubt.

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MAXIMUM DEVIATIONS ALLOWED ON ENDS OF EXTENSION SPRINGS



VARIATION IN ALIGNMENT OF ENDS (LOOPS)

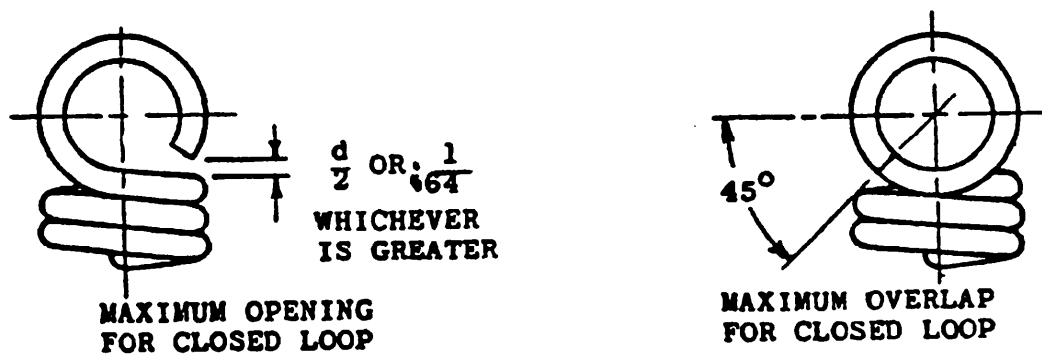


FIGURE 63

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materials are not ordinarily subject to hydrogen embrittlement during electroplating.

37. TOLERANCES FOR SPRINGS. Tables that follow present tolerances which can be obtained under normal manufacturing methods. Springs having closer tolerances can be produced but at greatly increased cost. A more liberal tolerance should be specified whenever possible.

TABLE XX. Tolerances on squareness of grinding on end coils for compression springs

Number of coils per inch of length (Number of coils $\div$ length)	Maximum tolerance from squareness to axis		
	Spring index (OD/d)		
	Up to 6 (small)	6 to 10 (average)	10 to 16 (large)
Up to 7	3°	4°	5°
Over 7 to 12	4°	5°	6°
Over 12 to 17	5°	6°	7°
Over 17	6°	7°	8°

These tolerances in degrees may be reduced 50 per cent by individually grinding, measuring, testing, and sorting the springs, but this increases production time and cost. Grinding end coils of springs made from wire under $\frac{1}{32}$ inch diameter is usually unneces-

sary and frequently is expensive.

37.1 TOLERANCE ON LOADS. The recommended tolerance on all loads is plus or minus 10 per cent and should be specified as shown on the drawing requirement charts. However, when the deflection from the free length to the first load is small, the load tolerance must be large and should be considered proportional to the tolerances that apply to the free length. For example, if the deflection is only $\frac{1}{8}$ inch and the tolerance on the free length is $\frac{1}{32}$ inch, then it follows that the tolerance on the first load should be plus or minus 25 per cent ($\frac{1}{32} = 25$ per cent of $\frac{1}{8}$), otherwise the tolerance on the free length could not be taken.

Where a specific amount of initial tension is required for extension springs, it should be specified in pounds or ounces with a recommended tolerance of plus or minus 15 per cent, but it usually is specified as a REFERENCE.

Smaller load tolerances than these usually require testing and adjusting each spring and add considerably to the manufacturing time and to the cost.

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TABLE XXI

Table of Third and Fourth Powers of Wire Diameters

d	d ³	d ⁴	d	d ³	d ⁴	d	d ³	d ⁴	d	d ³	d ⁴
.001	.000000001	.000000000001	.051	.00013265	.000073652	.101	.00103030	.00010406	.151	.00344295	.00051909
.002	.000000008	.00000000064	.052	.00014061	.000077316	.102	.00106121	.00010624	.152	.00351181	.00053379
.003	.000000027	.00000000243	.053	.00014888	.000081005	.103	.00109273	.00010855	.153	.0035816	.00054798
.004	.000000064	.00000000256	.054	.00015756	.000084831	.104	.00112486	.00011099	.154	.0036523	.00056245
.005	.000000125	.00000000625	.055	.00016658	.000088756	.105	.00115762	.00011355	.155	.0037239	.00057720
.006	.000000216	.00000001296	.056	.00017582	.000092835	.106	.00119102	.00011625	.156	.0037964	.00059224
.007	.000000343	.00000002401	.057	.00018519	.000096968	.107	.00122504	.00011908	.157	.0038699	.00060757
.008	.000000512	.00000004096	.058	.00019471	.000101165	.108	.00125971	.00012196	.158	.0039443	.00062320
.009	.000000729	.00000006561	.059	.00020438	.000105444	.109	.00129503	.00012486	.159	.0040197	.00063913
.010	.000001000	.00000010000	.060	.00021420	.000109800	.110	.00133100	.00012780	.160	.0040960	.00065536
.011	.000001331	.00000016771	.061	.00022416	.000114236	.111	.00136765	.00013081	.161	.0041733	.00067190
.012	.000001728	.000000220736	.062	.00023423	.000118743	.112	.00140499	.00013385	.162	.0042515	.00068875
.013	.000002197	.000000285001	.063	.00024440	.000123330	.113	.00144293	.00013690	.163	.0043307	.00070591
.014	.000002744	.000000360296	.064	.00025468	.000127887	.114	.00148148	.00014000	.164	.0044109	.00072329
.015	.000003375	.000000450625	.065	.00026507	.000132526	.115	.00152064	.00014314	.165	.0044921	.00074120
.016	.000004096	.000000553536	.066	.00027556	.000137247	.116	.00156040	.00014634	.166	.0045743	.00075933
.017	.000004913	.000000671321	.067	.00028615	.000142051	.117	.00160076	.00014959	.167	.0046575	.00077780
.018	.000005832	.000000808544	.068	.00029684	.000146930	.118	.00164173	.00015289	.168	.0047416	.00079659
.019	.000006859	.00000096661	.069	.00030763	.000151884	.119	.00168331	.00015623	.169	.0048268	.00081573
.020	.000008000	.00000116600	.070	.00031852	.000156910	.120	.00172550	.00015970	.170	.0049130	.00083521
.021	.000009261	.00000140851	.071	.00032951	.000162017	.121	.00176830	.00016321	.171	.0050002	.00085504
.022	.000010648	.000001696256	.072	.00034060	.000167206	.122	.00181171	.00016672	.172	.0050884	.00087521
.023	.000012167	.000002025001	.073	.00035179	.000172477	.123	.00185573	.00017023	.173	.0051777	.00089575
.024	.000013824	.00000240192	.074	.00036308	.000177830	.124	.00190036	.00017374	.174	.0052680	.00091664
.025	.000015625	.0000028246875	.075	.00037447	.000183265	.125	.00194560	.00017725	.175	.0053593	.00093789
.026	.000017576	.000003300296	.076	.00038596	.000188782	.126	.00199145	.00018086	.176	.0054516	.00095951
.027	.000019683	.000003829321	.077	.00039755	.000194381	.127	.00203791	.00018400	.177	.0055449	.00098151
.028	.000021952	.000004413056	.078	.00040924	.000200062	.128	.00208498	.00018714	.178	.0056392	.00100388
.029	.000024389	.000005052401	.079	.00042103	.000205825	.129	.00213266	.00019030	.179	.0057345	.00102663
.030	.000027000	.00000576000	.080	.00043292	.000211670	.130	.00218095	.00019356	.180	.0058308	.00104976
.031	.000029791	.000006537351	.081	.00044491	.000217697	.131	.00222985	.00019693	.181	.0059280	.00107328
.032	.000032768	.00000738528	.082	.00045700	.000223806	.132	.00227936	.00020040	.182	.0060262	.00109720
.033	.000035937	.00000830529	.083	.00046919	.000230097	.133	.00232948	.00020389	.183	.0061255	.00112151
.034	.000039304	.00000930896	.084	.00048148	.000236470	.134	.00238021	.00020740	.184	.0062258	.00114623
.035	.000042875	.00001039700	.085	.00049387	.000242925	.135	.00243155	.00021092	.185	.0063271	.00117135
.036	.000046656	.00001157120	.086	.00050636	.000249462	.136	.00248350	.00021445	.186	.0064294	.00119688
.037	.000050653	.00001283251	.087	.00051895	.000256081	.137	.00253605	.00021800	.187	.0065327	.00122283
.038	.000054872	.00001418288	.088	.00053164	.000262782	.138	.00258920	.00022156	.188	.0066370	.00124920
.039	.000059319	.00001562400	.089	.00054443	.000269565	.139	.00264295	.00022513	.189	.0067423	.00127599
.040	.000064000	.00001715840	.090	.00055732	.000276430	.140	.00269730	.00022870	.190	.0068486	.00130323
.041	.000068921	.00001878776	.091	.00057031	.000283377	.141	.00275225	.00023228	.191	.0069559	.00133099
.042	.000074088	.00002051312	.092	.00058340	.000290406	.142	.00280780	.00023587	.192	.0070642	.00135918
.043	.000079507	.00002233569	.093	.00059659	.000297517	.143	.00286395	.00023948	.193	.0071735	.00138780
.044	.000085184	.00002425744	.094	.00060988	.000304710	.144	.00292070	.00024310	.194	.0072838	.00141685
.045	.000091125	.00002628000	.095	.00062327	.000311985	.145	.00297805	.00024673	.195	.0073951	.00144629
.046	.000097336	.00002841456	.096	.00063676	.000319342	.146	.00303600	.00025037	.196	.0075074	.00147618
.047	.000103823	.00003066223	.097	.00065035	.000326781	.147	.00309455	.00025402	.197	.0076207	.00150651
.048	.000110592	.00003303408	.098	.00066404	.000334302	.148	.00315370	.00025768	.198	.0077350	.00153728
.049	.000117649	.00003553120	.099	.00067783	.000341905	.149	.00321345	.00026135	.199	.0078503	.00156852
.050	.000125000	.00003816000	.100	.00069172	.000349590	.150	.00327380	.00026503	.200	.0079666	.00160000

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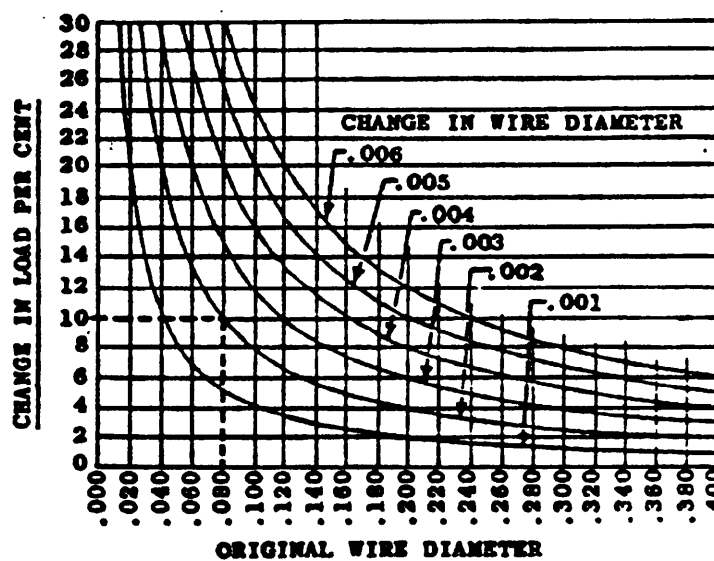
TABLE XXII. Load changes due to slight wire changes

INTRODUCTION: It is frequently necessary to make a slight change to the wire diameter of a spring, either to use a wire size available or to change the load to meet a specified requirement. Adding only .001 in. to .040 in. wire (thereby using .041 in. instead of .040 in.) increases the load 10 per cent. Adding .001 in. to $\frac{3}{16}$ in. wire, however, changes the load only 2 per cent, but a .005 in. addition increases the load 10 per cent. This data is for compression and extension springs.

CURVES: The curves below show the per cent change in loads due to slight changes in wire diameters. The curves are suitable for compression, extension and torsion springs using any material, with wire diameters between .015 in. and .375 in. inclusive.

EXAMPLE: If a spring that is supposed to be made from .080 in. wire, is actually made from .082 in. wire, the .002 in. heavier wire causes a heavier load. The per cent increase in load is 10 per cent as shown by the intersection of the vertical line .080 in. and the curve .002 in. (as shown dotted).

Similarly if .078 in. wire is used, the .002 in. lighter wire causes a lighter load of 10 per cent (same curve). If the original load must be maintained, it would be necessary to add 10 per cent more coils if the heavier wire is used or 10 per cent less coils with the lighter wire.



Note: Changing the wire size also changes the solid height, body length, stress, etc.

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TABLE XXIII. Causes of spring failure. Listed in sequence according to frequency of failure.

Spring failure may be caused by breakage, high permanent set or loss of load. Group I lists the causes that occur most frequently. Group II contains the less frequent causes and Group III lists causes that occur occasionally.

	Cause	Comment
GROUP I	HIGH STRESS	The majority of spring failures are due to high stresses caused by large deflections and high loads. High stresses should be used only for statically loaded springs. Low stresses lengthen fatigue life.
	HYDROGEN EMBRITTLEMENT	Improper electro-plating methods and acid cleaning of springs, without proper baking treatment, cause spring steels to become brittle, and is a frequent cause of failure. Non-ferrous springs are immune.
	SHARP BENDS & HOLES	Sharp bends on extension, torsion and flat springs and holes or notches in flat springs, cause high concentration of stress resulting in failure. Bend radii should be as large as possible, and tool marks avoided.
	FATIGUE	Repeated deflections of springs, especially above 1,000,000 operations, even with medium stresses, may cause failure. Low stresses should be used for severe operating conditions.
GROUP II	SHOCK LOADING	Impact, shock, and rapid loading cause far higher stresses than those computed by the regular spring formulas. High carbon spring steels do not withstand shock loading as well as alloy steels.
	CORROSION	Slight rusting or pitting caused by acids, alkali, galvanic corrosion, stress corrosion cracking, or corrosive atmosphere weakens the material and causes higher stresses in the corroded area.
	FAULTY HEAT TREATMENT	Keeping spring materials at the hardening temperature for longer periods than necessary causes an undesirable growth in grain structure resulting in brittleness even though the hardness may be correct.
	FAULTY MATERIAL	Poor material containing inclusions, seams, slivers, and flat material with rough, slit, or torn edges cause early failure. Overdrawn wire, improper hardness, and poor grain structure also result in early failure.
GROUP III	HIGH TEMPERATURE	High temperatures reduce spring temper (or hardness), lowers the modulus of elasticity thereby causing lower loads, reduces the elastic limit and increases corrosion. Corrosion resisting or nickel alloys should be used.
	LOW TEMPERATURE	Temperatures below — 40° F lessen the ability of carbon steels to withstand shock loads. Carbon steels become brittle at — 70° F. Corrosion resisting, nickel or non-ferrous alloys should be used.
	FRICTION	Close fits on rods or in holes result in a wearing away of material and occasional failure. The outside diameters of compression springs expand during deflection but they become smaller on torsion springs.
	OTHER CAUSES	Enlarged hooks on extension springs increases the stress at the bends. Carrying too much electrical current will cause failure. Welding and soldering frequently destroy the spring temper. Tool marks, nicks, and cuts often become stress raisers. Deflecting torsion springs outwardly causes high stresses. Winding them tightly causes binding on supporting rods. High speed of deflection, vibration and surging due to operation near natural periods of vibration or their harmonics, cause increased stresses.